

Project Management in the IT Context

SECTION: Why IT Projects Are Different

IT projects tend to fail more often than projects in many other industries because of characteristics specific to software and technology work:

- **Intangibility:** software cannot be seen or touched the way a bridge or building can, making progress harder to verify visually.
- **Rapidly changing technology:** requirements and platforms can shift mid-project, invalidating earlier decisions.
- **Complexity and interdependency:** a change in one module can ripple unpredictably through a codebase or system architecture.
- **Requirements volatility:** business needs (and therefore scope) often change faster in IT than in construction or manufacturing.

SECTION: The CHAOS Report Context

Industry surveys (most famously the Standish Group's CHAOS studies) have repeatedly found that a large share of IT projects are "challenged" (late, over budget, or reduced in scope) or fail outright. Commonly cited causes include:

- Lack of user involvement
- Unclear or incomplete requirements
- Unrealistic timeframes
- Lack of executive/sponsor support
- Poor change management

SECTION: IT Project Life Cycle Models

- **Predictive (Waterfall/Plan-driven):** scope is defined up front, phases (requirements → design → build → test → deploy) run sequentially. Suits well-understood, stable requirements (e.g. regulatory compliance systems).
- **Iterative/Incremental:** the system is built in repeated cycles, refining requirements as understanding grows.
- **Adaptive (Agile):** short iterations, continuous stakeholder feedback, embraces changing requirements even late in development.
- **Hybrid:** combines predictive planning (e.g. for infrastructure or fixed-budget contracts) with agile execution (e.g. for the software layer).

SECTION: Systems Development Life Cycle (SDLC) Alignment

IT project managers usually align the project life cycle with an SDLC model such as:

1. Planning
2. Analysis (requirements gathering)
3. Design
4. Implementation (coding, testing)
5. Support/maintenance

Understanding the SDLC in use affects how the PM plans scope, schedule, and risk — e.g. Agile SDLCs need rolling-wave planning rather than a single detailed schedule up front.

SECTION: Special IT Risk Factors

- **Technical risk:** unproven technology, integration with legacy systems.
- **Scope creep:** stakeholders requesting "just one more feature" is especially common in software because changes feel deceptively easy to make.
- **Vendor/tooling risk:** dependency on third-party APIs, libraries, or cloud services that may change or become deprecated.
- **Security and data privacy risk:** increasingly central given regulations like GDPR.

SECTION: The Role of the IT Project Manager

Beyond generic PM duties, an IT PM typically needs to:

- Translate business requirements into technical specifications understandable to developers.
- Manage technical debt trade-offs (short-term speed vs long-term maintainability).
- Coordinate with architects, QA leads, and DevOps/operations teams.
- Manage stakeholder expectations around what "done" means for a digital product (e.g. MVP vs full feature set).

SECTION: Common Exam Angles

- Why do IT projects have higher failure rates than construction projects? (intangibility, requirements volatility, technology change)
- Compare predictive vs adaptive life cycles and when each is appropriate.
- Given a scenario, identify whether a failure mode is a scope, schedule, technical, or stakeholder-management problem.